IoT devices and network devices (e.g., router, hub	Main considerations	Specific questions	Responses
4 × Arduino Opta RS485 zone controllers; AAEON BOXER-8653AI industrial edge computer; TP-Link managed PoE+ switch; local operator web application; maintenance log/API.	*Determine the connectivity between IoT devices and network devices in the edge/fog layer*	*Is an edge/fog layer required? Why?*	Yes. Deterministic control must survive Internet loss, and 20 continuous video streams are too bandwidth-intensive and privacy-sensitive for routine cloud upload. Opta PLCs form the device edge; the BOXER is the farm fog node for protocol normalisation, quality checks, video inference, event correlation, short-term storage and store-and-forward delivery.
		Which network device(s) should handle computing across the edge/fog layer?	Each Opta executes five-tank control rules and exposes states over the isolated control VLAN. The AAEON BOXER-8653AI runs containerised acquisition, message brokering, rules, local dashboards and vision models. The managed switch and router forward and secure traffic but do not make animal-control decisions.
		What data is transmitted between the IoT devices and network device(s)?	Probes send temperature, pH, turbidity, ammonia, level, flow, feed distance and humidity with timestamps and diagnostics. PLCs add set-points, relay states, accumulated flow, manual/automatic mode, watchdog and alarm codes. Cameras send H.265 RTSP substreams and event windows. The edge adds device identity, tank/zone context and quality flags.
		What data processing and/or analysis tasks should be performed in the edge/fog layer?	PLCs scale signals, debounce pulses, check plausible ranges/rates, enforce hysteresis and time/volume bounds, and execute fail-safe control. The BOXER synchronises timestamps, detects stale or contradictory readings, aggregates summaries, correlates flow with level/valve state, infers activity/feeding/nesting candidates, queues messages during outages and retains rolling evidence locally.
		If the fog connects with a user application, what is the purpose of the connection?	On the farm LAN, authorised users see current tank status, local alarms, event video, acknowledgement state and bounded manual controls even during WAN loss. The fog node labels unsynchronised data and preserves operational visibility and intervention capability.
		If the fog connects with a system, what is the purpose of the connection?	The fog service writes an append-only operations log and exports alarms, acknowledgements, maintenance flags, calibration due dates and actuator runtime through signed CSV/API. The maintenance system can create or close work orders but cannot issue raw heater, pump or valve commands.

IoT devices and network devices (e.g., router, hub	Main considerations	Specific questions	Responses
AAEON BOXER-8653AI edge computer; Teltonika RUTX11 firewall/router; AWS IoT Core, time-series/object storage and rules; role-based web application; maintenance/work-order system.	*Determine the connectivity to the cloud*	*Is a cloud layer required? Why?*	Yes, but not for immediate safety. AWS provides durable off-site history, multi-user access, fleet identity, long-term comparison, reporting and notification integration. AWS IoT Core authenticates the edge through X.509 certificates, routes MQTT messages and maintains desired/reported state through device shadows (Amazon Web Services, n.d.).
		What data should be transmitted to the cloud, and is it from IoT devices or edge/fog layer devices?	The BOXER publishes validated minute summaries, state changes, alarms, acknowledgements, health/calibration metadata and daily aggregates by MQTT/TLS. Selected event thumbnails/clips and diagnostic bundles use HTTPS object upload. Raw continuous camera streams and high-frequency samples remain on site unless an authorised investigation requests them.
		What data processing and/or analysis tasks should be performed in the cloud?	IoT rules route telemetry to time-series storage and alarms to notification workflows; object storage retains approved evidence clips. Cloud jobs calculate cross-unit baselines, growth-stage comparisons, sensor-drift indicators and feed/energy/water summaries. Device shadows retain configuration state, while audit records capture user changes and acknowledgements.
		If the cloud connects with a user application, what is the purpose of the connection?	A role-based web application provides historical charts, unit comparison, alarm notifications, evidence links, weekly reports and audit history. Users may request bounded set-point changes; the cloud records the request, but the edge/PLC rejects stale, unauthorised or out-of-range values.
		If the cloud connects with a system, what is the purpose of the connection?	The cloud exports approved maintenance tickets, calibration schedules, equipment runtime, water/feed consumption and compliance summaries through an API or scheduled CSV. Only necessary identifiers and evidence are included. Return status can close a ticket, but the external system cannot issue raw actuator commands.